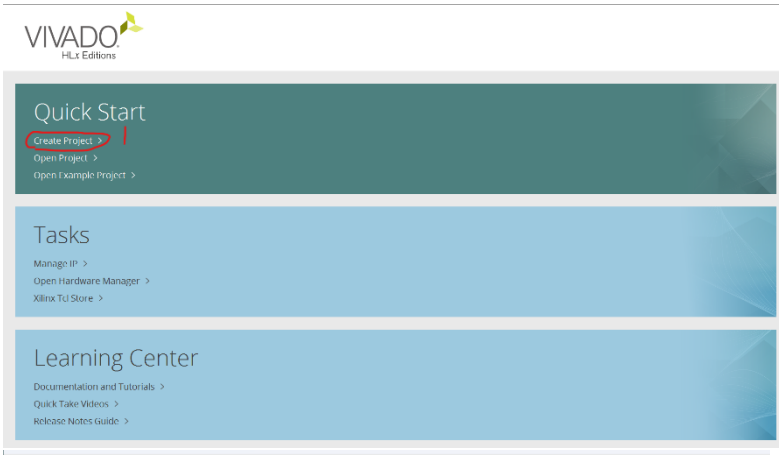
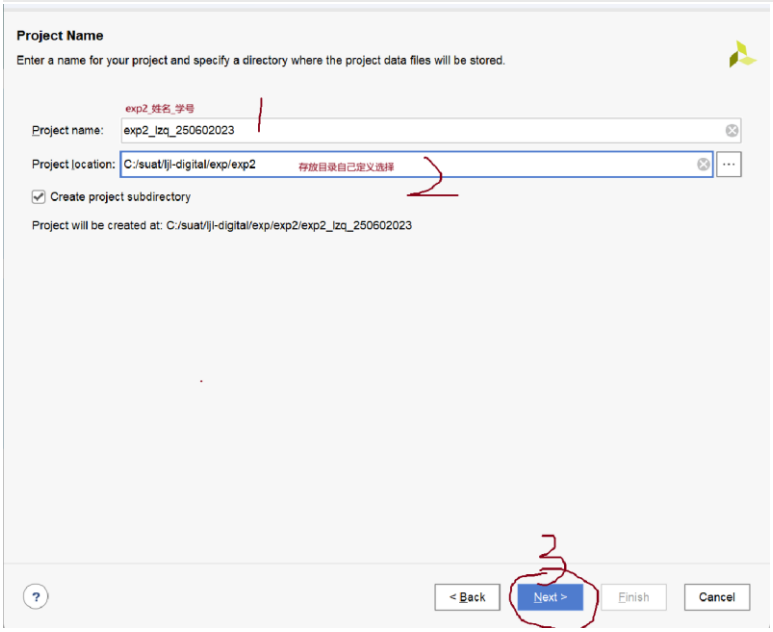
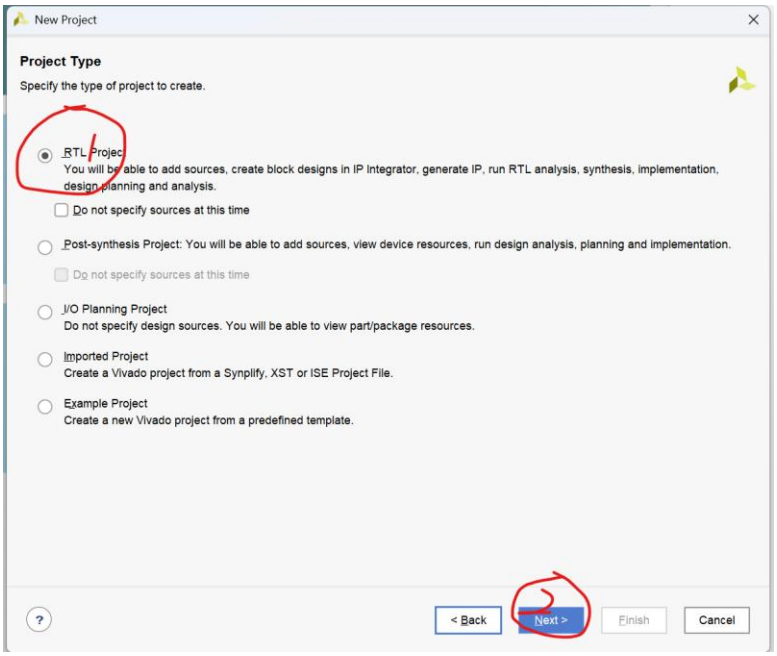
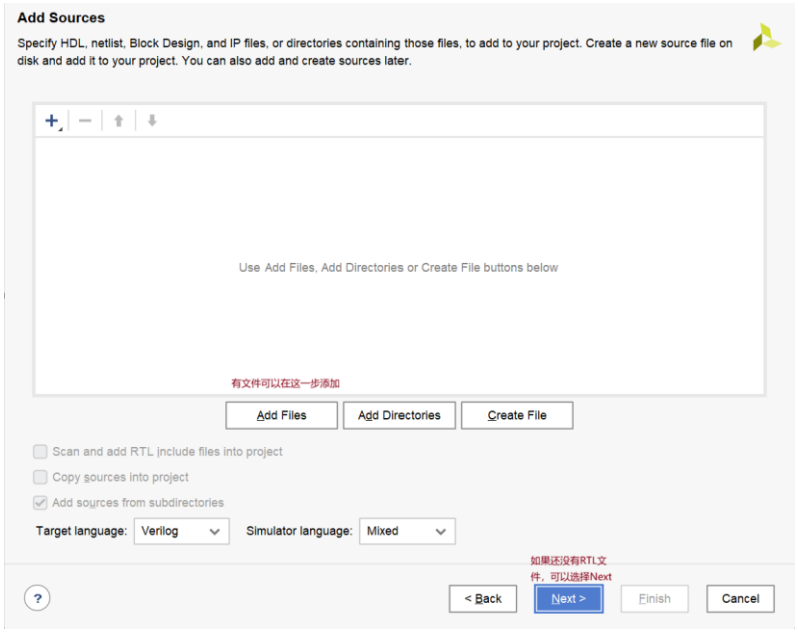
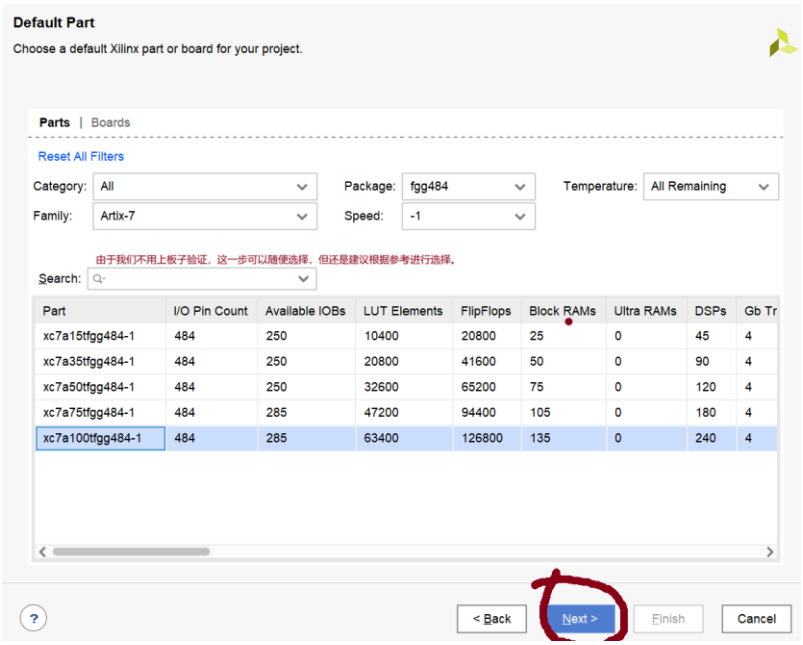
1、Vivado 使用教程

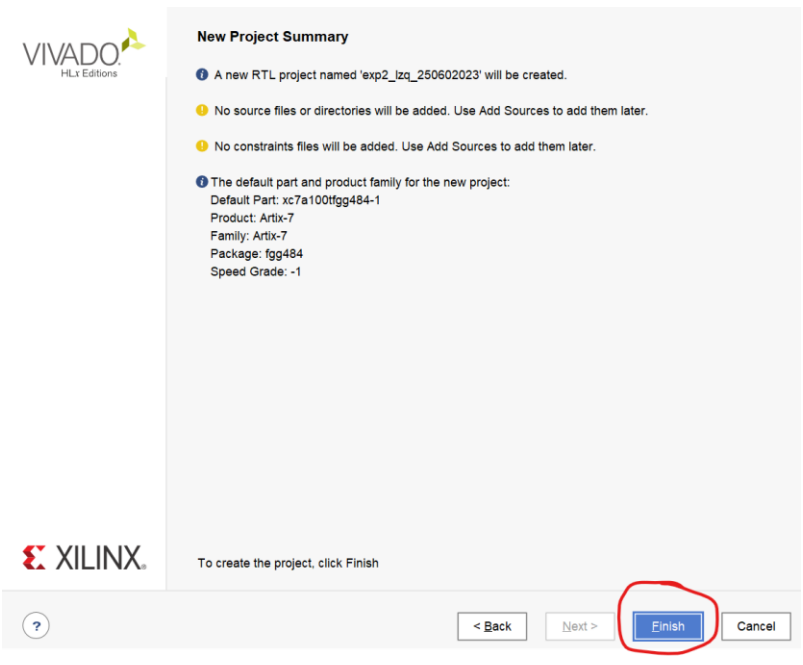


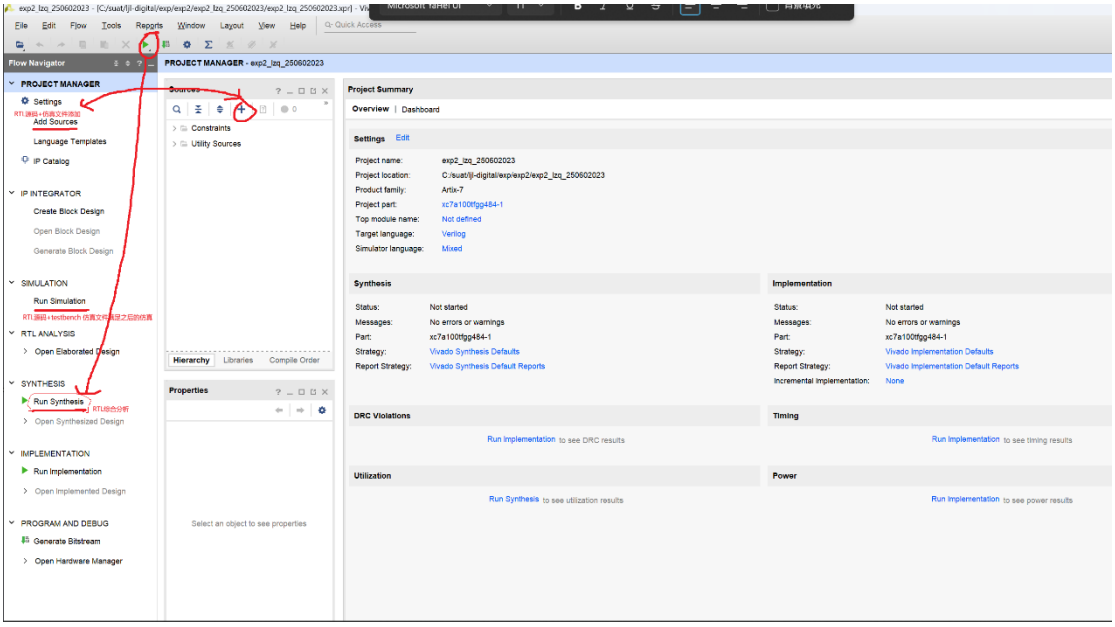




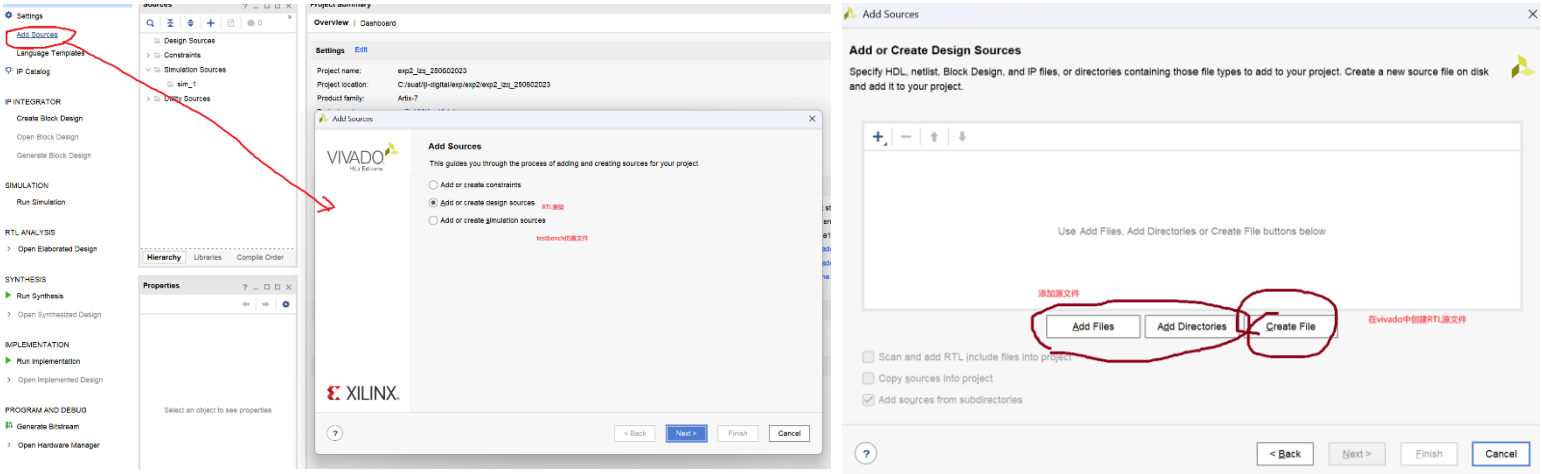

Add Sources
Specify HDL, netlist, Block Design, and IP files, or directories containing those files, to add to your project. Create a new source file on disk and add it to your project. You can also add and create sources later.
Use Add Files, Add Directories or Create File buttons below
有文件可以在这一步添加
Add Files
Add Directories
Create File
Scan and add RTL include files into project
Copy sources into project
Add sources from subdirectories
Target language: Verilog
Simulator language: Mixed
如果还没有RTL文件，可以选择Next
< Back
Next >
Finish
Cancel


Default Part
Choose a default Xilinx part or board for your project.
Parts | Boards
Reset All Filters
Category: All
Package: fgg484
Temperature: All Remaining
Family: Artix-7
Speed: -1
由于我们不用上板子验证，这一步可以随便选择，但还是建议根据参考进行选择。
Search:
Part | I/O Pin Count | Available IOBs | LUT Elements | FlipFlops | Block RAMs | Ultra RAMs | DSPs | Gb Tr
xc7a15tfgg484-1 | 484 | 250 | 10400 | 20800 | 25 | 0 | 45 | 4
xc7a35tfgg484-1 | 484 | 250 | 20800 | 41600 | 50 | 0 | 90 | 4
xc7a50tfgg484-1 | 484 | 250 | 32600 | 65200 | 75 | 0 | 120 | 4
xc7a75tfgg484-1 | 484 | 285 | 47200 | 94400 | 105 | 0 | 180 | 4
xc7a100tfgg484-1 | 484 | 285 | 63400 | 126800 | 135 | 0 | 240 | 4
< Back
Next >
Finish
Cancel


VIVADO HLx Editions
New Project Summary
A new RTL project named 'exp2_lzq_250602023' will be created.
No source files or directories will be added. Use Add Sources to add them later.
No constraints files will be added. Use Add Sources to add them later.
The default part and product family for the new project:
Default Part: xc7a100tfgg484-1
Product: Artix-7
Family: Artix-7
Package: fgg484
Speed Grade: -1
XILINX
To create the project, click Finish
< Back
Next >
Finish
Cancel

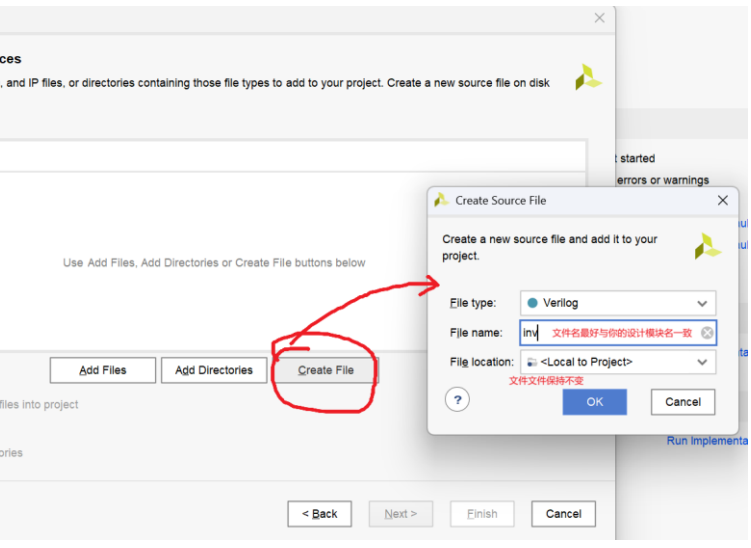


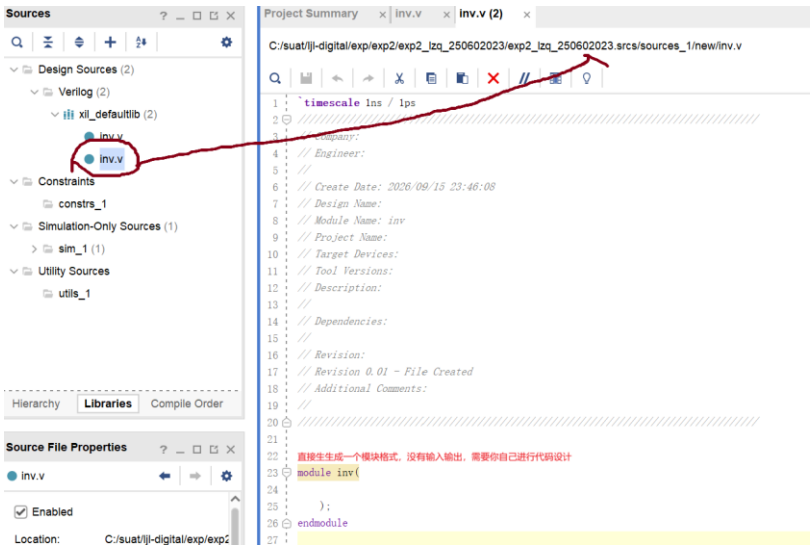
2、进入项目界面之后，开始添加 RTL 源码+testbench 文件，也可以在 vivado 中编写



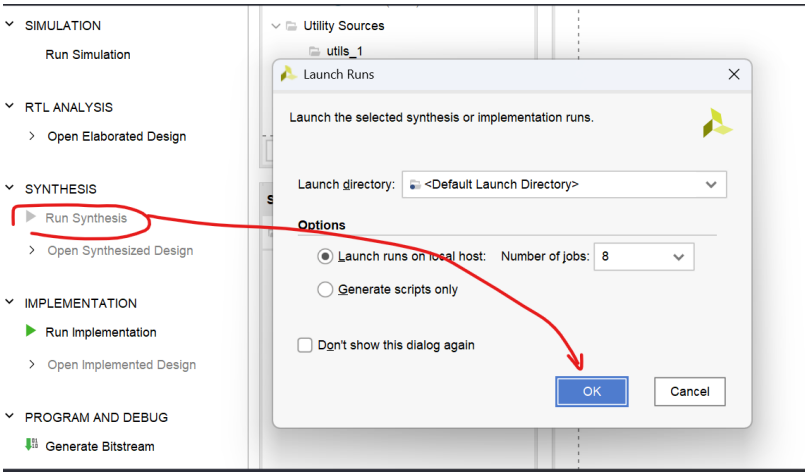
添加文件就直接把你的 RTL 设计文件和 testbench，simulation 文件提交上传即可。

下图是 vivado 中创建文件的流程：

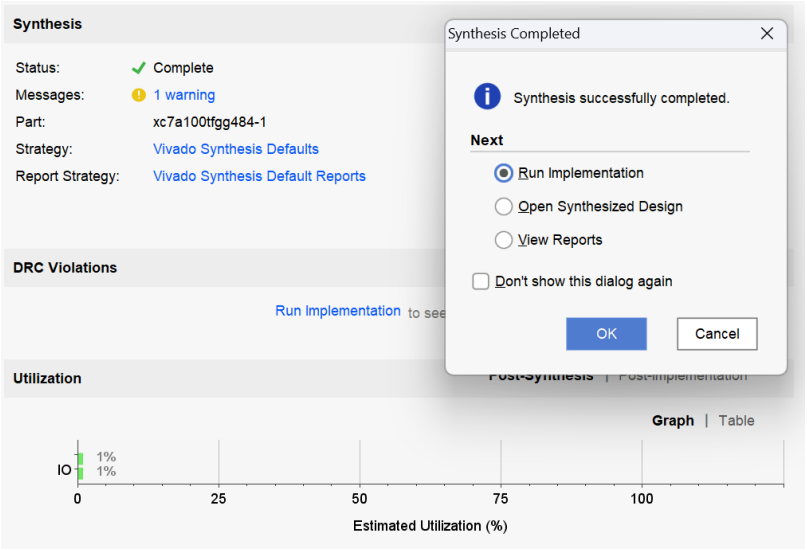




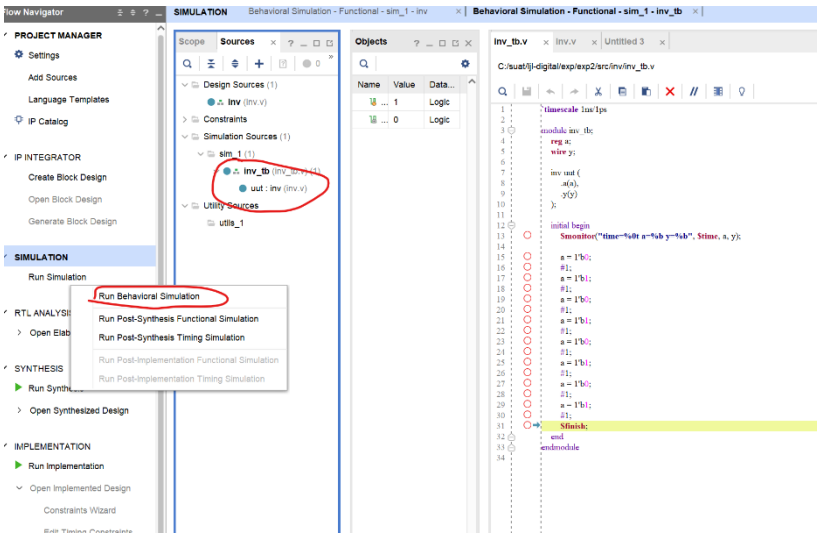
3、将 RTL 源码和 testbench 文件都搞定后，进行先进行综合，再执行仿真

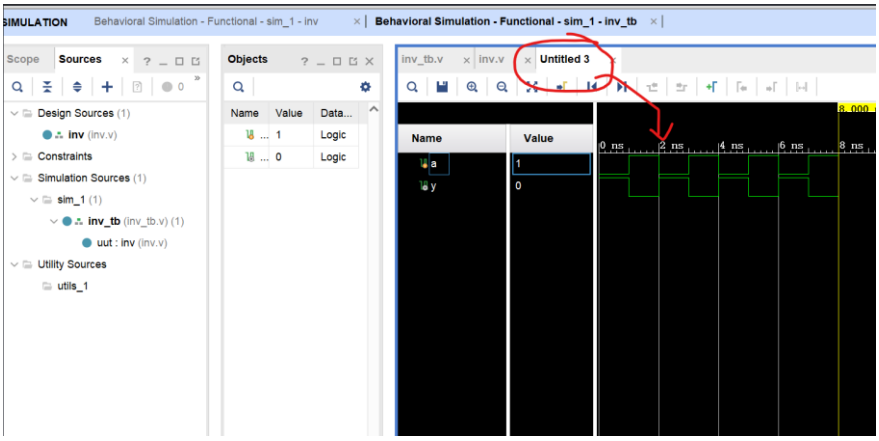


等待几之后出现这个界面，综合成功，就可以取消，run implement 属于布局布线执行



3、接下来是执行波形仿真：





以上就是整个 vivado 简单的仿真流程，核心主要是对软件的熟悉使用，然后就是代码的编写正确。软件的使用可以进行网络搜素，或者 AI 查询。

课堂练习：基于以下图示完成源代码的编写以及 testbench 文件（波形仿真文件）

对硬件 verilog，FPGA 开发，IC 设计的感兴趣的同学想加强 verilog 语言代码的学习理解。最基础的 verilog 在线编写网页交互式仿真学习网址如下
https://hdlbits.01xz.net/wiki/Main_Page

课堂练习 1;

图 1-1，为基础的与门，请根据图示结构编写 RTL 以及 testbench 仿真文件完成 vivado 的波形仿真。

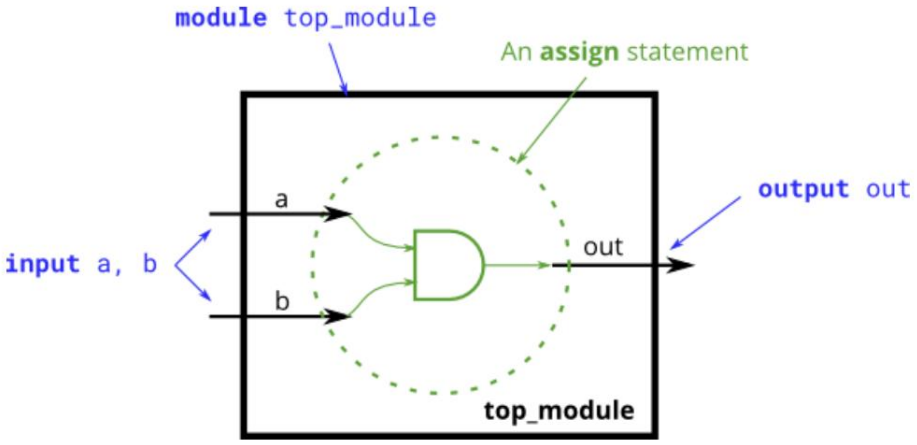


图 1-1

Testbench 源码参考如下:

```
`timescale 1ns / 1ps

module tb_and;

    reg  a, b;
    wire out;

    top_module uut (
        .a(a),
        .b(b),
        .out(out)
    );

    initial begin
        a = 0; b = 0; #10;
        a = 0; b = 1; #10;
        a = 1; b = 0; #10;
        a = 1; b = 1; #10;

        $finish;
    end

endmodule
```

课堂练习 2;

图 2-1，为基础的或门，请根据图示结构编写 RTL 以及 testbench 仿真文件完成 vivado 的波形仿真。

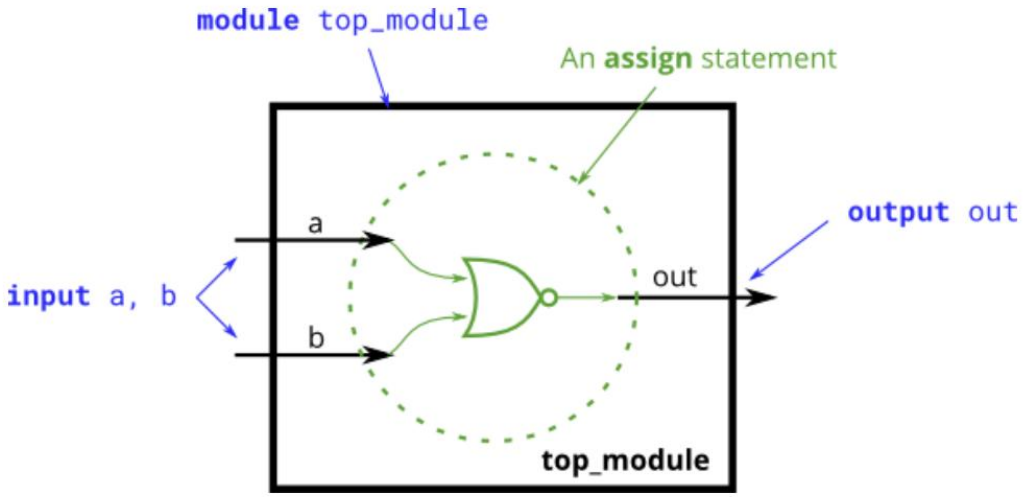


图 2-1

Testbench 源码参考如下:

```
`timescale 1ns / 1ps

module tb_or;

    reg  a, b;
    wire out;

    top_module uut (
        .a(a),
        .b(b),
        .out(out)
    );

    initial begin
        a = 0; b = 0; #10;
        a = 0; b = 1; #10;
        a = 1; b = 0; #10;
        a = 1; b = 1; #10;

        $finish;
    end

endmodule
```

课后作业 1;

图 3-1，为基础的异或门，请根据图示结构编写 RTL 以及 testbench 仿真文件完成 vivado 的波形仿真。

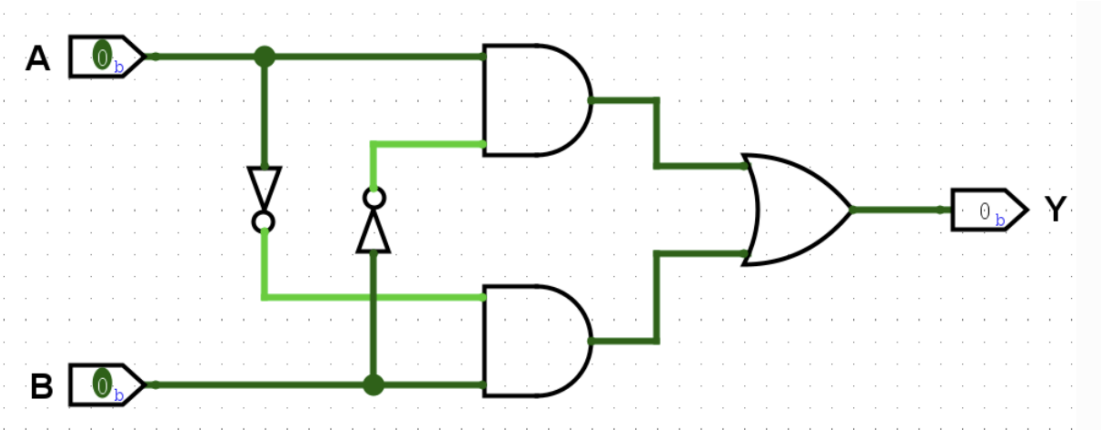


图 3-1

Testbench 源码参考如下:

```
`timescale 1ns / 1ps

module tb_xor;

    reg  A, B;
    wire Y;

    top_module uut (
        .A(A),
        .B(B),
        .Y(Y)
    );

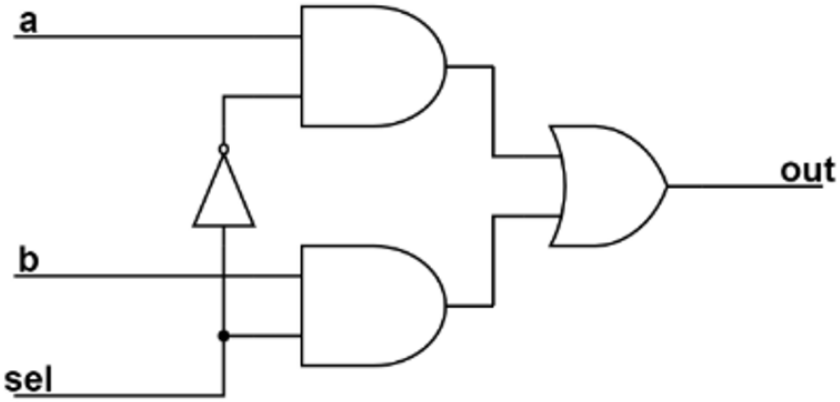
    initial begin
        A = 0; B = 0; #10;
        A = 0; B = 1; #10;
        A = 1; B = 0; #10;
        A = 1; B = 1; #10;

        $finish;
    end

endmodule
```

课后作业 2;

图 3-1，为基础的选择器，请根据图示结构编写 RTL 以及 testbench 仿真文件完成 vivado 的波形仿真。



Testbench 源码参考如下:

```
`timescale 1ns / 1ps

module tb_mux;

    reg  a, b, sel;
    wire out;

    top_module uut (
        .a(a),
        .b(b),
        .sel(sel),
        .out(out)
    );

    initial begin
        sel = 0; a = 0; b = 0; #10;
        sel = 0; a = 0; b = 1; #10;
        sel = 0; a = 1; b = 0; #10;
        sel = 0; a = 1; b = 1; #10;

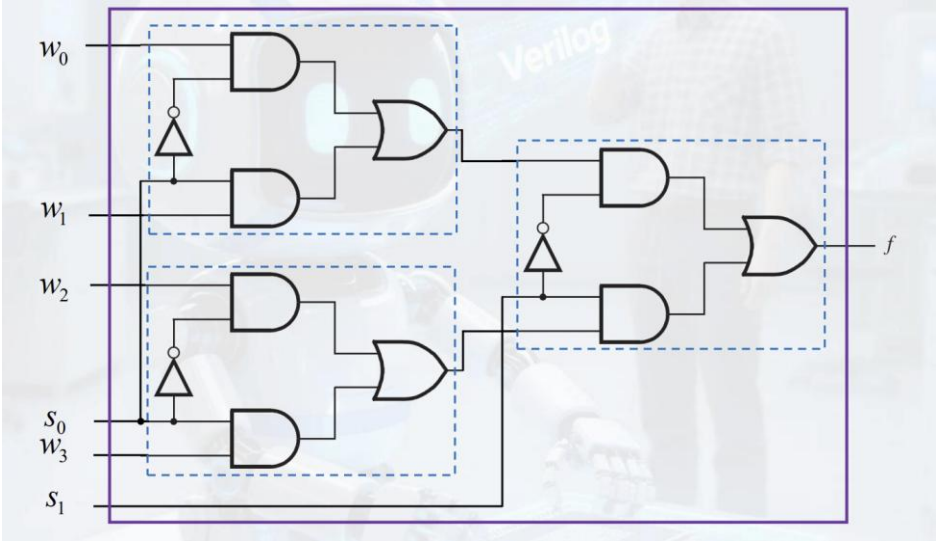
        sel = 1; a = 0; b = 0; #10;
        sel = 1; a = 0; b = 1; #10;
        sel = 1; a = 1; b = 0; #10;
        sel = 1; a = 1; b = 1; #10;

        $finish;
    end

endmodule
```

课后作业 2;

图 4-1，基于 1 位多路 2 选一选择器扩展为 2 路二选一多路选择器，请根据图示结构编写 RTL 以及 testbench 仿真文件完成 vivado 的波形仿真



Testbench 源码参考如下：

```
`timescale 1ns / 1ps

module tb_mux4;

    reg  w0, w1, w2, w3;
    reg  s0, s1;
    wire f;

    integer i;

    top_module uut (
        .w0(w0),
        .w1(w1),
        .w2(w2),
        .w3(w3),
        .s0(s0),
        .s1(s1),
        .f(f)
    );

    initial begin
        // 遍历 6 个输入的全部 64 种组合
        for (i = 0; i < 64; i = i + 1) begin
            {w3, w2, w1, w0, s1, s0} = i;
            #10;
        end

        $finish;
    end

endmodule
```

仿真结果波形，缩放至 640ns 波形大致如图所示，供给给参考。

